Problem 12.4

Brian and Jennifer each take out a loan of X. Jennifer will repay her loan by making one payment of 800 at the end of year 10. Brian will repay his loan by making one payment of 1,120 at the end of year 10. The nominal semiannual rate being charged to Jennifer is exactly one-half the nominal semiannual rate being charged to Brian. Calculate X.

Solution

Let $j^{(2)}$ be Brian's nominal annual interest rate convertible semiannually. Jennifer's nominal annual rate convertible semiannually is then $j^{(2)}/2$. Define $k = j^{(2)}/4$. There are 20 semiannual periods in 10 years, so the accumulation factors are $(1+2k)^{20}$ for Brian and $(1+k)^{20}$ for Jennifer.

Since each loan has the same present value X,

$$\begin{aligned}
\frac{1{,}120}{(1+2k)^{20}} &= \frac{800}{(1+k)^{20}}, \\
\left(\frac{1+2k}{1+k}\right)^{20} &= \frac{1{,}120}{800} = 1.4.
\end{aligned}$$

Let $r = 1.4^{1/20}$. Then

$$\begin{aligned}
\frac{1+2k}{1+k} &= r, \\
1+2k &= r(1+k), \\
k &= \frac{r-1}{2-r} \\
&\approx 0.017258736.
\end{aligned}$$

Using Jennifer's repayment to determine the loan amount,

$$\begin{aligned}
X &= \frac{800}{(1+k)^{20}} \\
&\approx 568.147921.
\end{aligned}$$

Therefore, each loan amount is approximately

$$\boxed{X \approx 568.15}.$$